

CE 310 — Week 4 Assignment: Poisson Probabilities and Return Periods

Due: Wednesday of Week 5 by 9:00 AM — submit to D2L

Overview

This assignment extends the Week 4 lab into new Poisson scenarios. You will compute event probabilities at thresholds and time horizons not covered in the lab, interpret those results through an engineering design lens, and explain in writing what the Poisson model reveals that a simple average cannot.

All three problems use the same dataset: Week4_Tucson_Precipitation_Daily_2013_2022.csv

How to Submit

- ☐ Following this submission format exactly is essential — with a class this size, grading depends on it. Submissions that don't comply will be penalized.

Requirement	Detail
File format	.xlsx only — written responses go in the Answer Sheet cells, not in a separate PDF
Sheet name	Must be exactly Answers — do not rename or delete it
Major declaration	Enter CE or ArcE in cell B4 — this determines which 2b row is graded
Column layout	Enter your formula in Column C and the resulting value in Column D — do not insert, delete, or move columns
Formulas	Type them as text — put an apostrophe before the =, e.g. '=COUNTIF(B2:B8761,">90")', so Excel stores the formula instead of running it. Screenshots are not accepted
Written responses	Type them in the yellow cells lower down the same sheet — do not submit a separate PDF
File name	CE310_W4_<NetID>.xlsx (e.g. CE310_W4_abc123.xlsx)

Problem 1 — New Poisson Probabilities (30 pts)

The lab computed λ for Summer, Spring, Winter, and July. This problem asks you to compute three probabilities not found in the lab — using a new threshold, a new season, and a new time horizon.

Part 1a — An Extreme July (10 pts)

July's $\lambda = 10.3$ rain days per month. Use this rate to compute:

$$P(X \geq 12 \mid \lambda_{\text{july}})$$

The probability that a July will have 12 or more rain days — roughly a 40% increase over the mean.

Note: Use your lab B6 answer for λ_{july} . Do not re-derive it from scratch.

Show your formula and the result. In 1–2 sentences, interpret this value: is a 12-rain-day July a rare exception or a reasonably common outcome?

Enter your response in the P1a_written cell of your Answer Sheet.

Part 1b — A Dry Summer Month (10 pts)

The lab computed $P(X = 0 \mid \lambda_{\text{spring}})$ and $P(X = 0 \mid \lambda_{\text{winter}})$ but not for Summer. Compute:

$$P(X = 0 \mid \lambda_{\text{summer}})$$

The probability that a Summer month (June, July, or August) has zero rain days.

Note: Use your lab A7 answer for λ_{summer} .

Show your formula and result. In 2–3 sentences, compare this to the $P(\text{dry spring month})$ from lab B3. Explain physically why the probability of a dry summer month is so much lower than a dry spring month in Tucson.

Enter your response in the P1b_written cell of your Answer Sheet.

Part 1c — A Very Dry Spring (10 pts)

Compute:

$$P(X \leq 2 \mid \lambda_{\text{spring}})$$

The probability that a Spring month has 2 or fewer rain days — a noticeably dry spring.

Note: Use your lab B2 answer for λ_{spring} .

Show your formula and result. In 1–2 sentences, explain what the CDF value $P(X \leq 2)$ tells you about the typical spring month in Tucson.

Enter your response in the P1c_written cell of your Answer Sheet.

Problem 2 — Seasonal Rain Rate and Building Envelope (18 pts)

Your lab computed four seasonal λ values. This problem uses the Fall rate — a season the lab did not analyze in depth — and connects it to building envelope performance.

2a — both tracks. Compute Fall lambda from the dataset:

Report the Fall rain day count (in the P2a_count row) and λ_{fall} (in the P2a_lambda row). Fall is September, October and November across all 10 years — 30 season-months in total, so $\lambda_{\text{fall}} = (\text{rain days}) \div 30$.

In 1–2 sentences, place Fall in context relative to your four seasonal λ values: which season is wettest, which is driest, and where does Fall rank?

Enter your response in the P2a_written cell of your Answer Sheet.

2b — answer ONLY the version matching your major. Declare CE or ArcE in cell B4 of your Answer Sheet; that declaration determines which of the two rows below is graded. Leave the other blank.

2b (Civil Engineering). Municipal drainage infrastructure is specified against a return period rather than a monthly rate. A culvert is designed for the 100-year storm — an event with a 1% chance of occurring in any given year — and the structure is expected to remain in service for a 50-year design life.

Compute the probability that at least one 100-year storm occurs during that 50-year design life:

$$P(\text{at least one}) = 1 - e^{-t/T}$$

where $t = 50$ years (design life) and $T = 100$ years (return period).

Note: In Excel this is $= (1 - \text{EXP}(-50/100)) * 100$. This is the Poisson zero-event complement — the same structure as $P(X = 0)$ in Part 1b, applied to a time

horizon instead of a month. Most students expect the answer to be 50%; it is not, and that gap is the point.

CE students: enter this in the P2b_CE row of your Answer Sheet.

2b (Architectural Engineering). Building envelope performance depends on how often moisture exposure repeats within a season, not on a once-a-century design event. For the Fall season, compute:

$$P(X \geq 5 \mid \lambda_{\text{fall}})$$

The probability that a Fall month has 5 or more rain days — a level at which prolonged moisture exposure becomes a concern for cladding, flashing, and membrane performance.

Note: Use $= (1 - \text{POISSON.DIST}(4, \lambda_{\text{fall}}, \text{TRUE})) * 100$. Note the 4, not the 5:
 $P(X \geq 5) = 1 - P(X \leq 4)$.

ArcE students: enter this in the P2b_ArcE row of your Answer Sheet.

Show your formula and report the result.

2c. In 3–4 sentences, synthesize your own 2b result with the seasonal λ table from Chart 2 of the lab. Explain why an engineer should use a Poisson exceedance probability rather than simply citing the annual average rain days. Reference ASHRAE 90.1-2022's requirement to size drainage systems for design-storm conditions. CE students: frame the recommendation around sizing a culvert whose service life spans many return periods. ArcE students: frame it around specifying window flashing, roofing membrane, and exterior cladding.

Enter your response in the P2c_written cell of your Answer Sheet.

Problem 3 — Reflection (12 pts)

Answer in one paragraph (100–150 words):

A building owner reviews your Week 4 lab and says: “Tucson gets about 3.9 rain days per month on average, so I’ll size my roof drains for 4 rain days per month. That should cover it.”

Using at least one specific number from your lab or this assignment, explain why this reasoning is flawed. Your response should address what $P(X > 4 \mid \lambda_{\text{overall}})$ tells you, and what the seasonal Poisson analysis reveals that the annual average obscures.

Enter your response in the P3_written cell of your Answer Sheet.

Submission Requirements

- Type all written responses in the designated yellow cells of your Assignment Answer Sheet. Submit CE310_W4_<NetID>.xlsx to D2L.
 - Show all Excel formulas with the formula bar visible for each POISSON.DIST or EXP expression.
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Grading Summary

Problem	Points
Problem 1a — $P(X \geq 12 \mid \text{July})$: formula, result, 1–2 sentence interpretation	10
Problem 1b — $P(X = 0 \mid \text{Summer})$: formula, result, 2–3 sentence comparison to spring	10
Problem 1c — $P(X \leq 2 \mid \text{Spring})$: formula, result, 1–2 sentence CDF interpretation	10
Problem 2 — Seasonal Rain Rate and Building Envelope (2a count + λ_{fall} + written, 2b by major, 2c written)	18
Problem 3 — Reflection paragraph (100–150 words, at least one specific number cited)	12
Total	60 pts

This assignment is worth 60 raw points and will be scaled to the Post-Lab Memo category in the course gradebook.